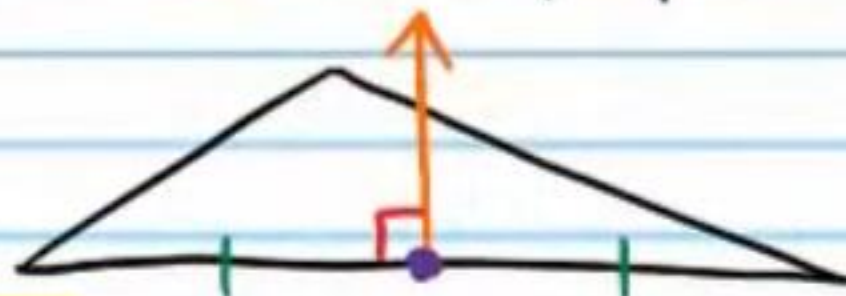


Points of Concurrency Generated by Triangles

Special Lines, Rays, and Segments of Triangles

Perpendicular Bisector: A line, ray, or segment that bisects a side of a triangle and is perpendicular to it. There are three perpendicular bisectors in each triangle.

Example:



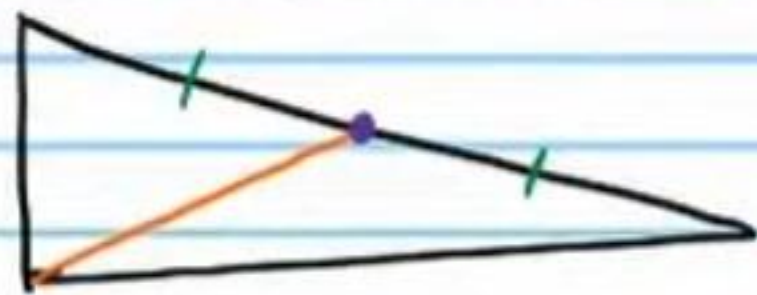
Angle Bisector: A line, ray, or segment that bisects an inner angle of a triangle. There are three angle bisectors in each triangle.

Example:

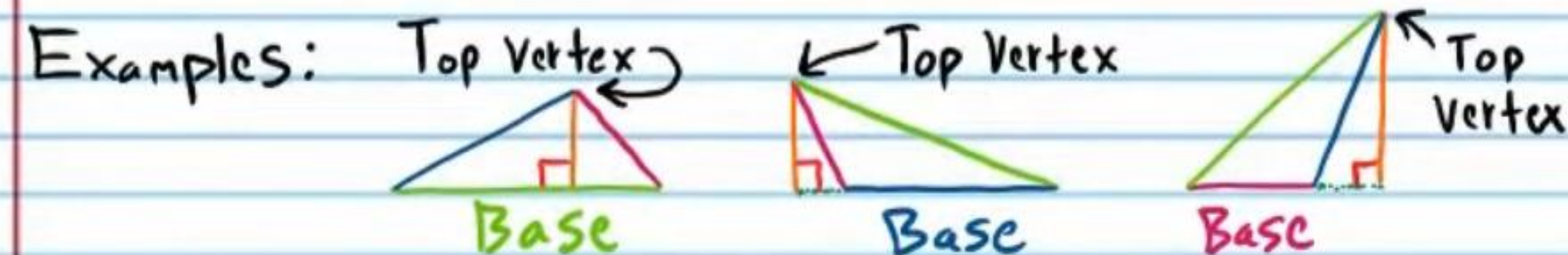


Median: A segment that extends from a vertex of a triangle to the midpoint of the opposite side. There are three medians in each triangle.

Example:

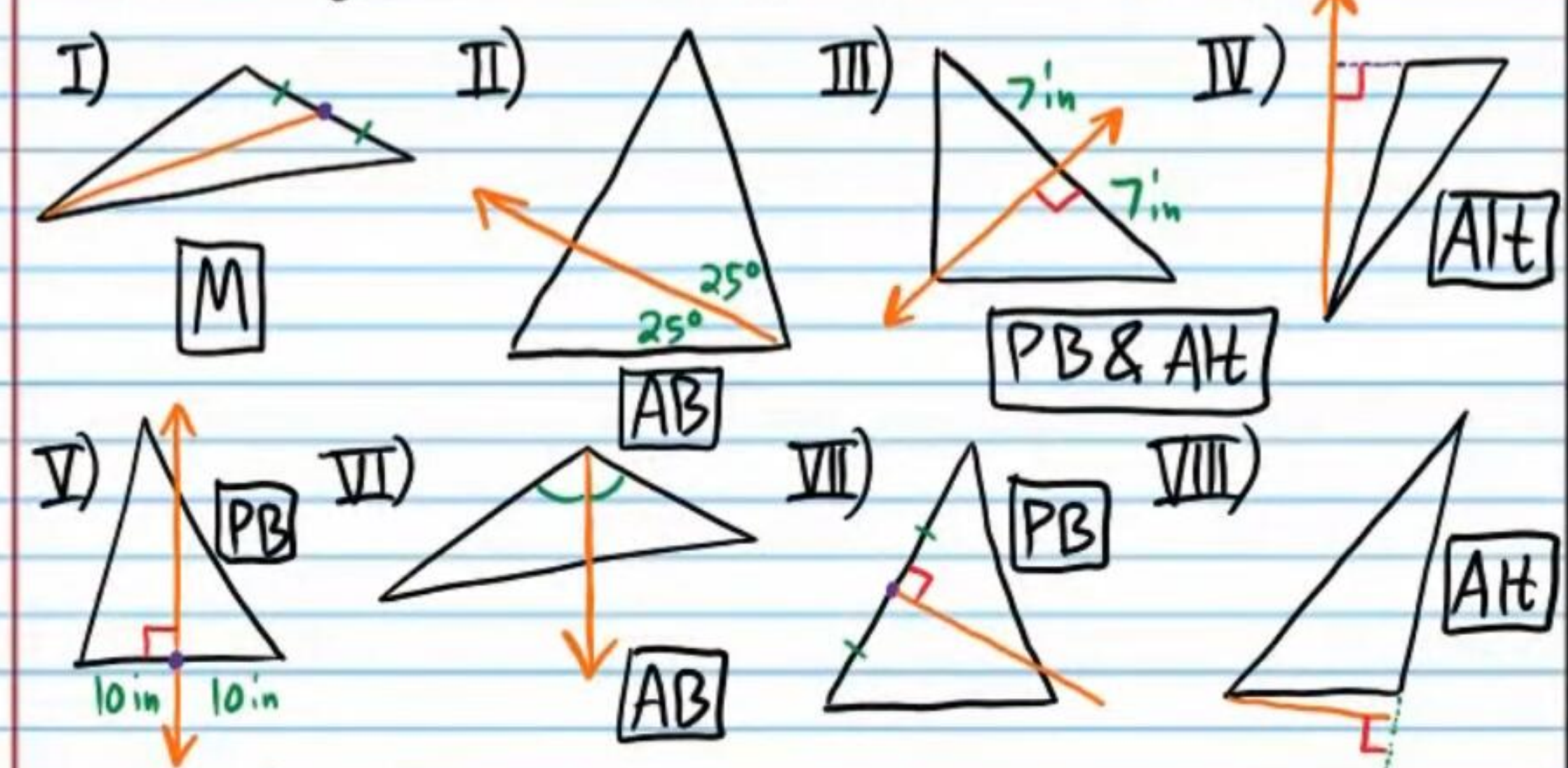


Altitude: A segment that represents the height of a triangle if the triangle rests on a particular side. The altitude extends from the top vertex and is perpendicular to the base (i.e. the opposite side). There are three altitudes in each triangle.

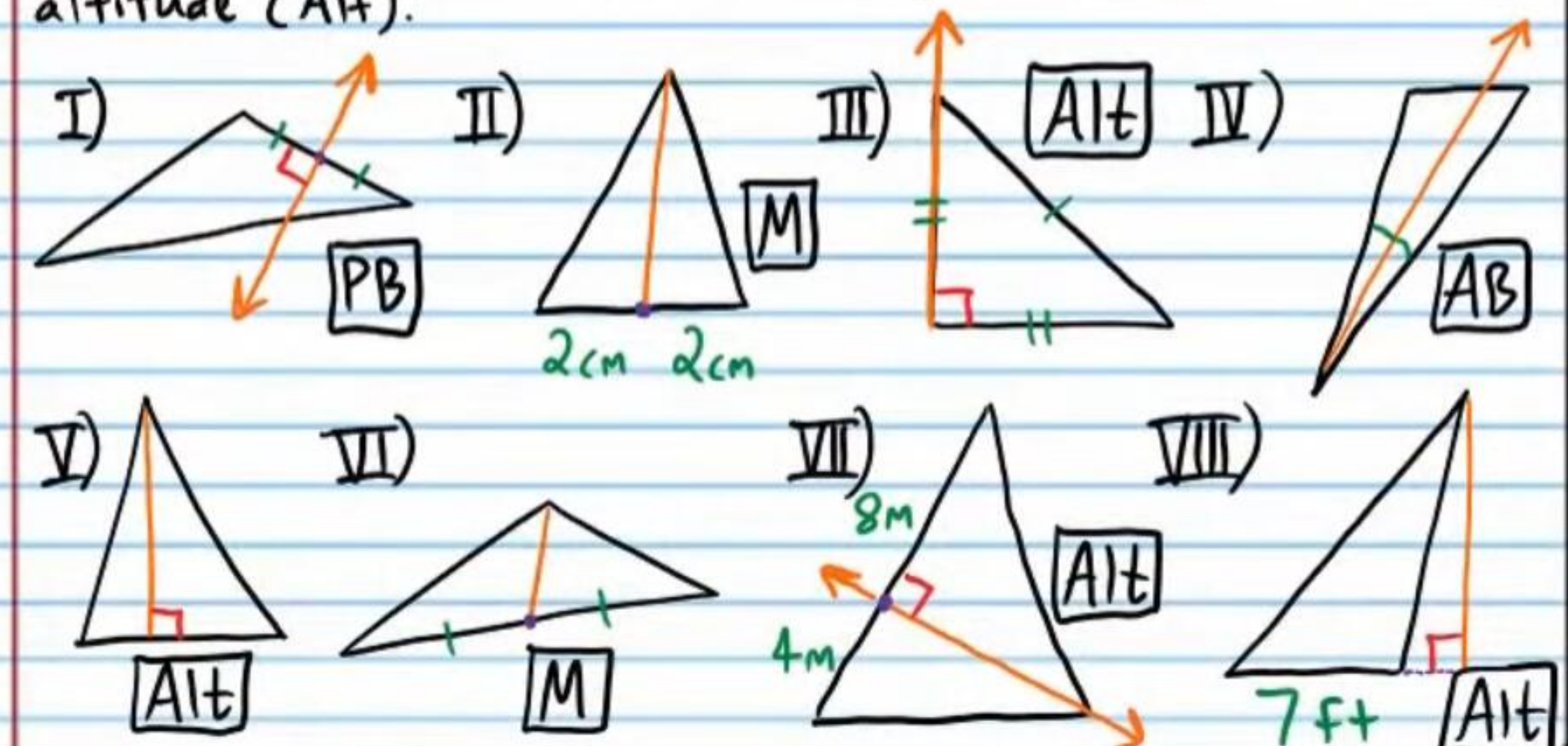


Practice

① Identify the line, ray, or segment as a perpendicular bisector (PB), angle bisector (AB), median (M), or line (or ray or segment) containing the altitude (Alt).

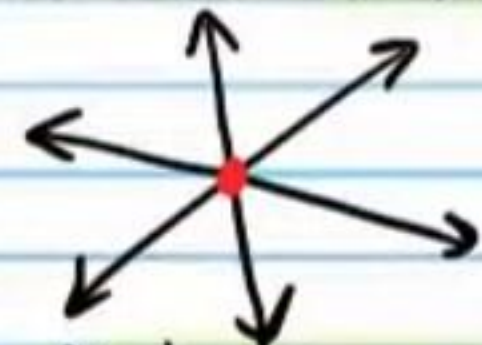


② Identify the line, ray, or segment as a perpendicular bisector (PB), angle bisector (AB), median (M), or line (or ray or segment) containing the altitude (Alt).

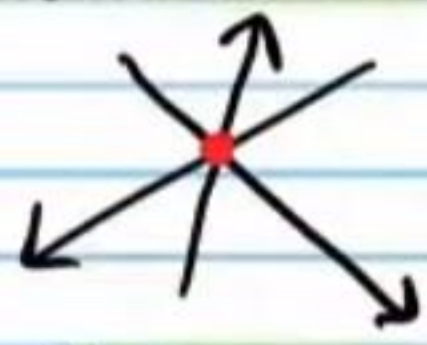
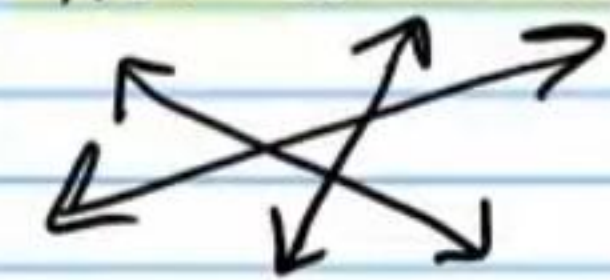


Concurrent Lines/Rays/Segments: When three or more lines/rays/segments intersect at a common point.

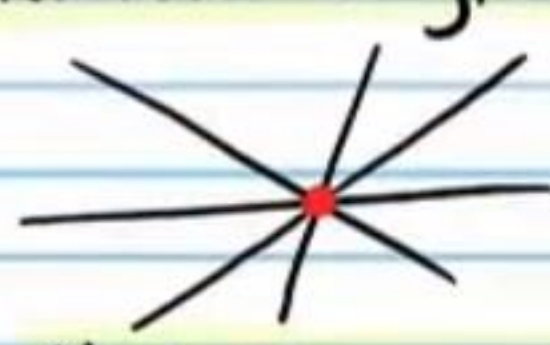
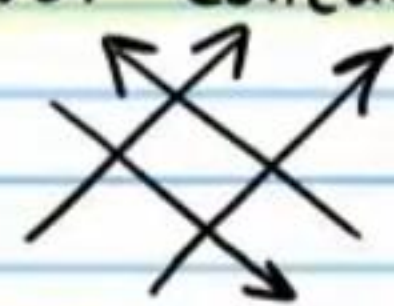
Concurrent Lines Concurrent Rays Concurrent Segments



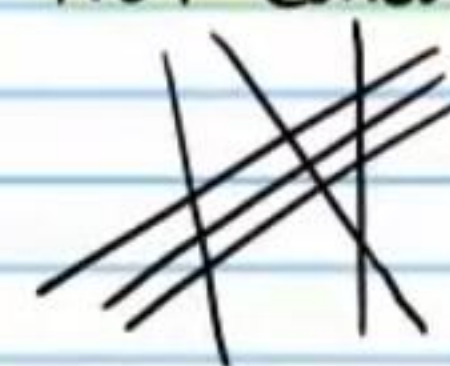
Not Concurrent



Not Concurrent



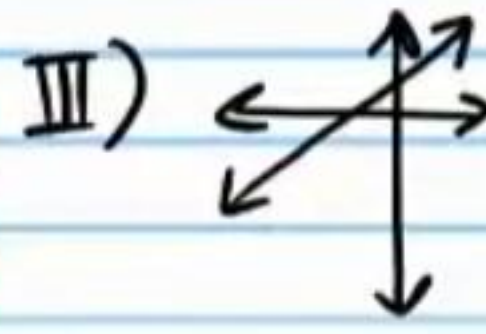
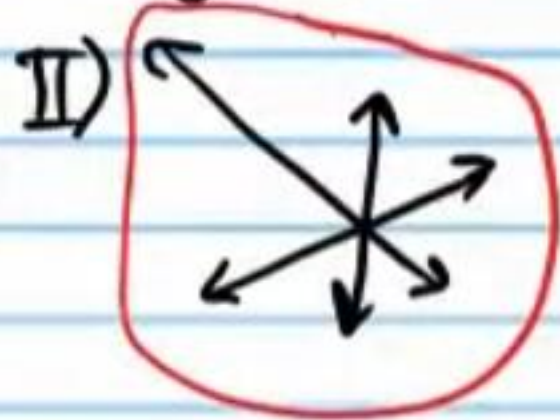
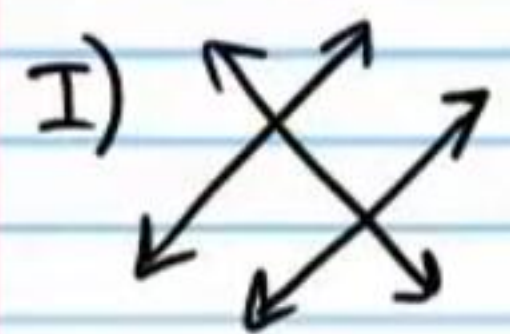
Not Concurrent



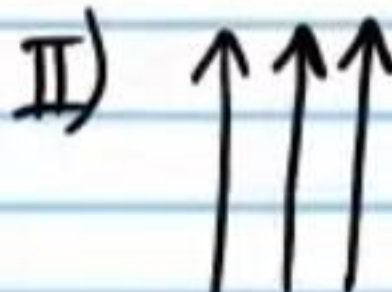
• **Point of Concurrence:** Where three or more lines, rays, or segments intersect.

Practice

③ Which lines are concurrent? You may select more than one group.



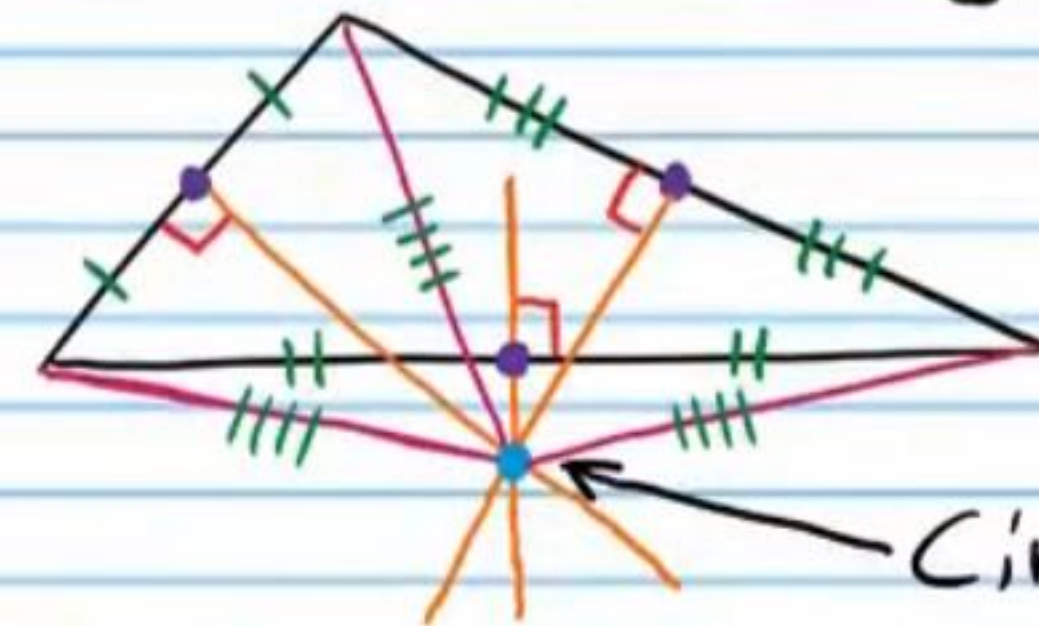
④ Which rays are concurrent? You may select more than one group.



Points of Concurrence Generated by Special Lines/Rays/Segments of Triangles

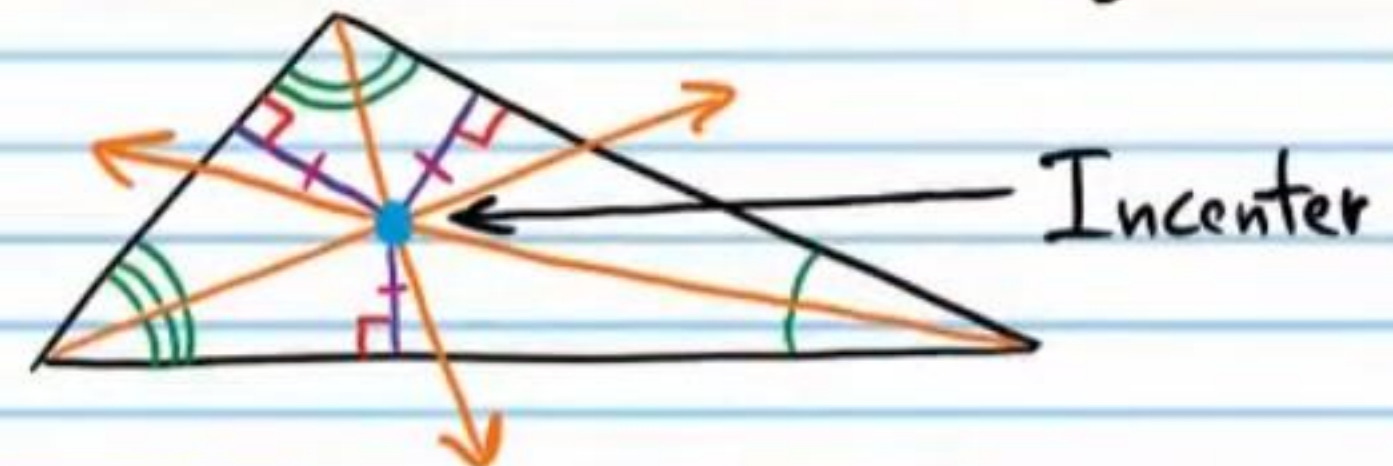
Circumcenter Theorem: The perpendicular bisectors of a triangle are concurrent and the point of concurrency (called the circumcenter) is equidistant from the vertices of the triangle.

Example:



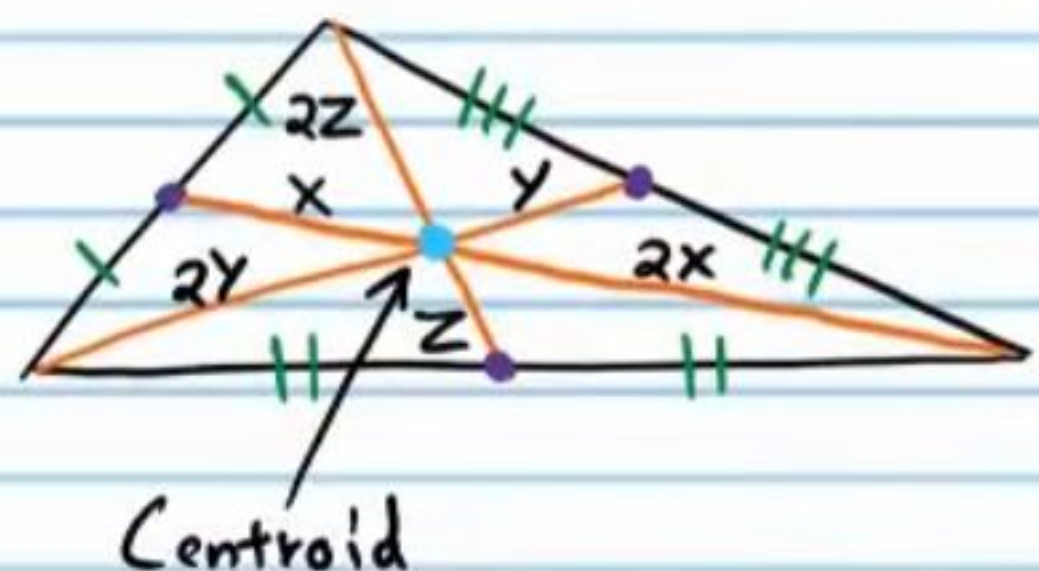
Incenter Theorem: The angle bisectors of a triangle are concurrent and the point of concurrency (called the incenter) is always inside the triangle and equidistant from the sides of the triangle.

Example:



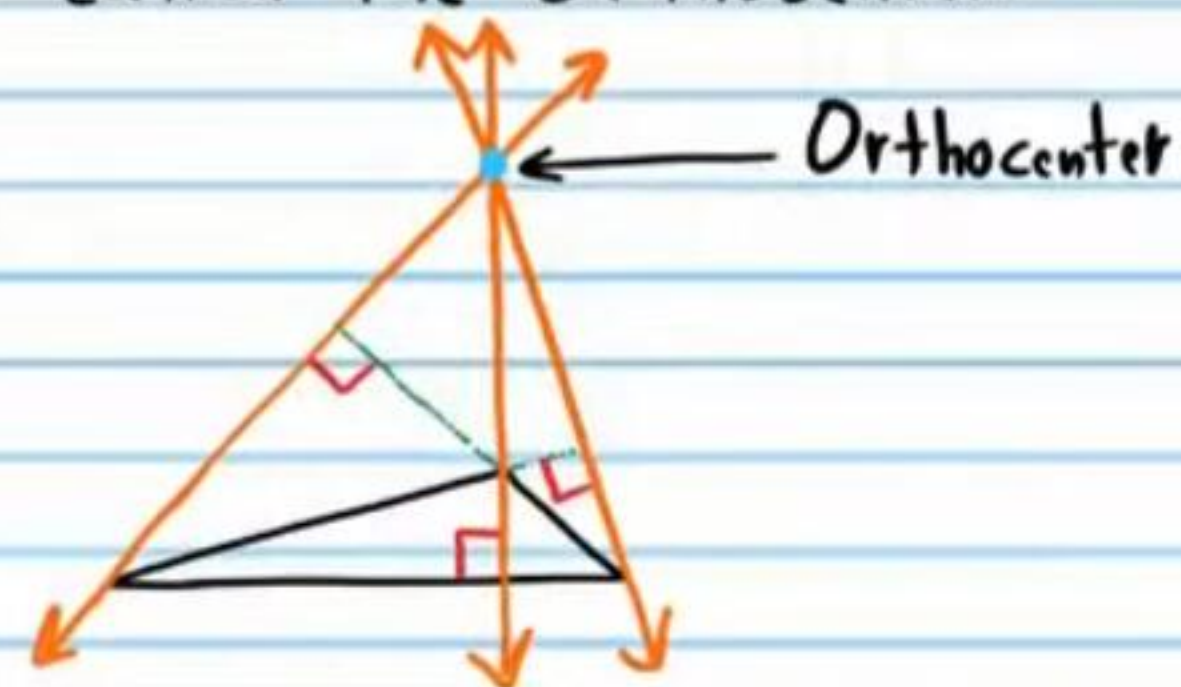
Centroid Theorem: The medians of a triangle are concurrent and the point of concurrency (called the centroid) is always inside the triangle and $\frac{2}{3}$ the distance from each vertex to the opposite side.

Example:



Orthocenter Theorem: The three lines containing the altitudes of a triangle are concurrent and the point of concurrency is called the orthocenter.

Example:



Practice

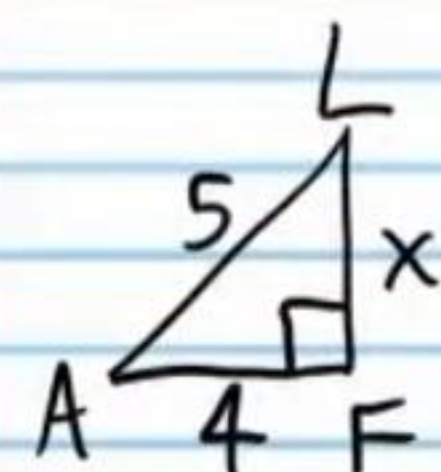
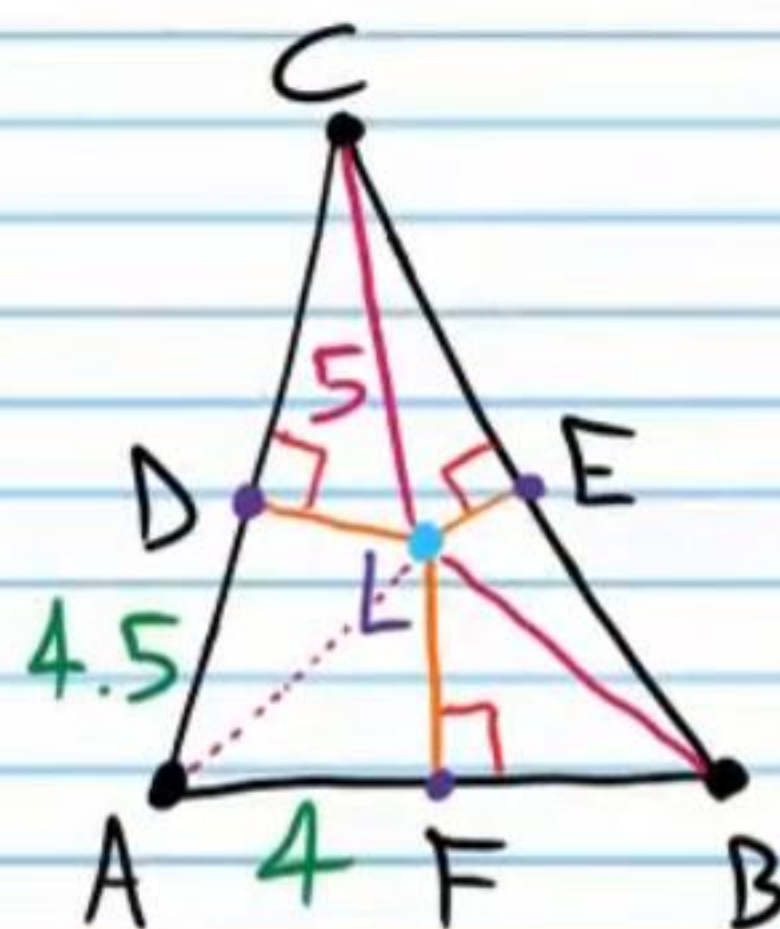
If point L is the circumcenter of each triangle below, find the quantities listed.

⑤ a) $CD = \boxed{4.5}$

b) $FB = \boxed{4}$

c) $AL = \boxed{5}$

d) $LF = \boxed{3}$



$$a^2 + b^2 = c^2$$

$$4^2 + x^2 = 5^2$$

$$16 + x^2 = 25$$

$$x^2 = 9$$

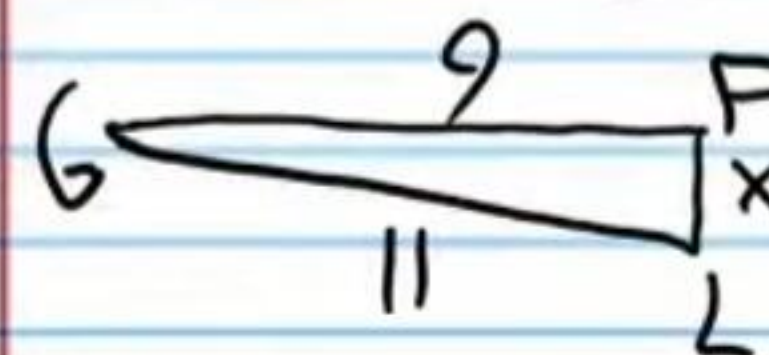
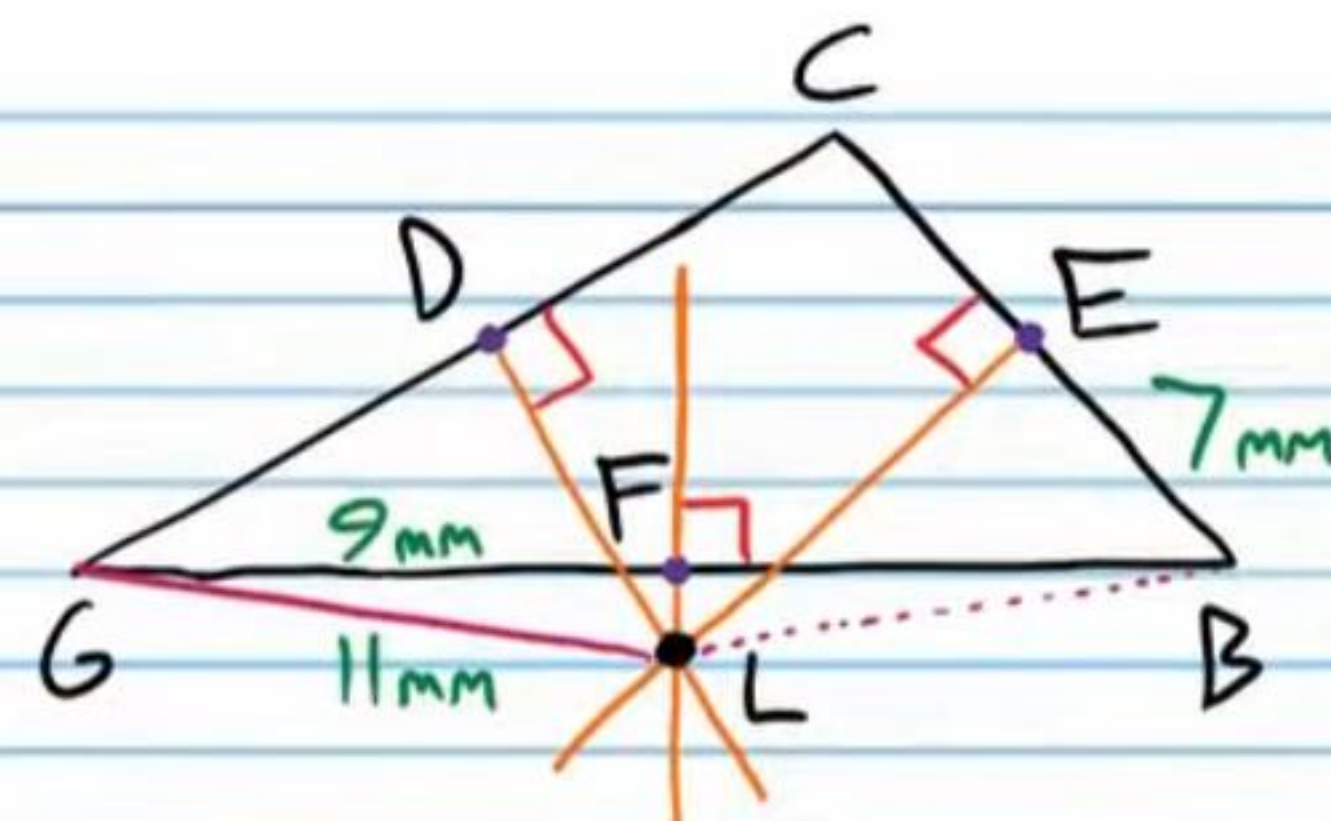
$$x = 3$$

⑥ a) $FB = \boxed{9\text{ mm}}$

b) $CE = \boxed{7\text{ mm}}$

c) $LB = \boxed{11\text{ mm}}$

d) $FL = \boxed{2\sqrt{10}\text{ mm}}$



$$9^2 + x^2 = 11^2$$

$$81 + x^2 = 121$$

$$x^2 = 40$$

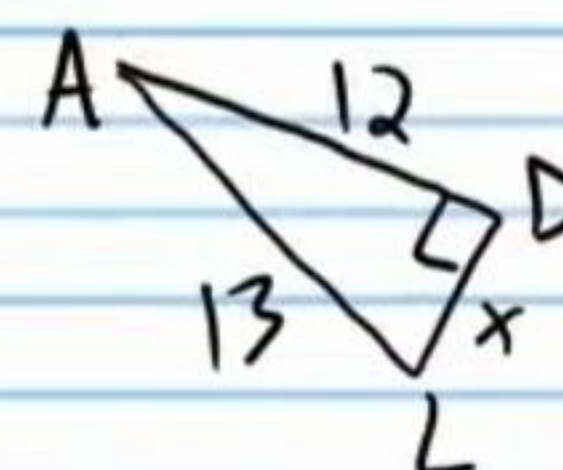
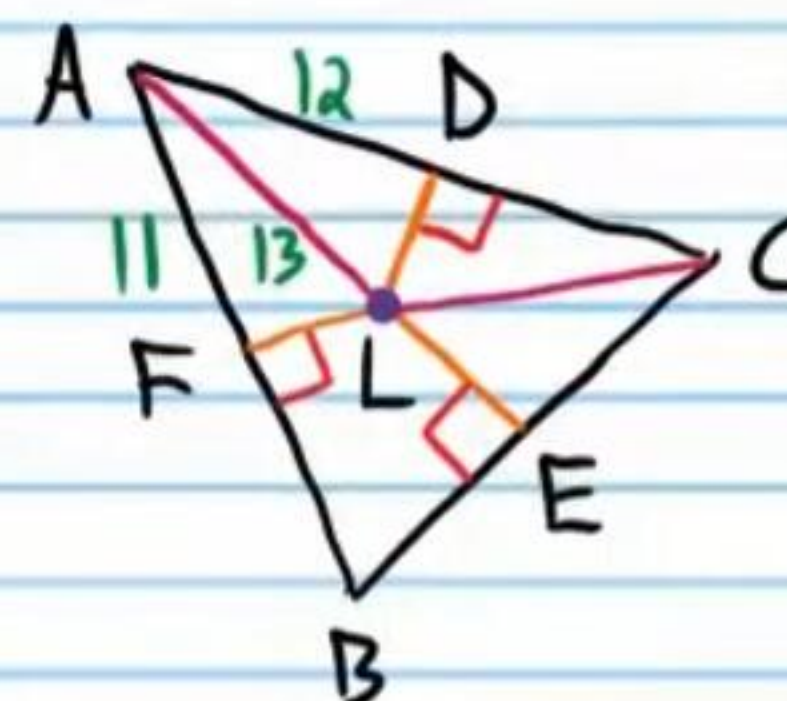
$$x = \sqrt{40} = 2\sqrt{10}$$

⑦ a) $DC = \boxed{12}$

b) $LC = \boxed{13}$

c) $FB = \boxed{11}$

d) $LD = \boxed{5}$



$$12^2 + x^2 = 13^2$$

$$144 + x^2 = 169$$

$$x^2 = 25$$

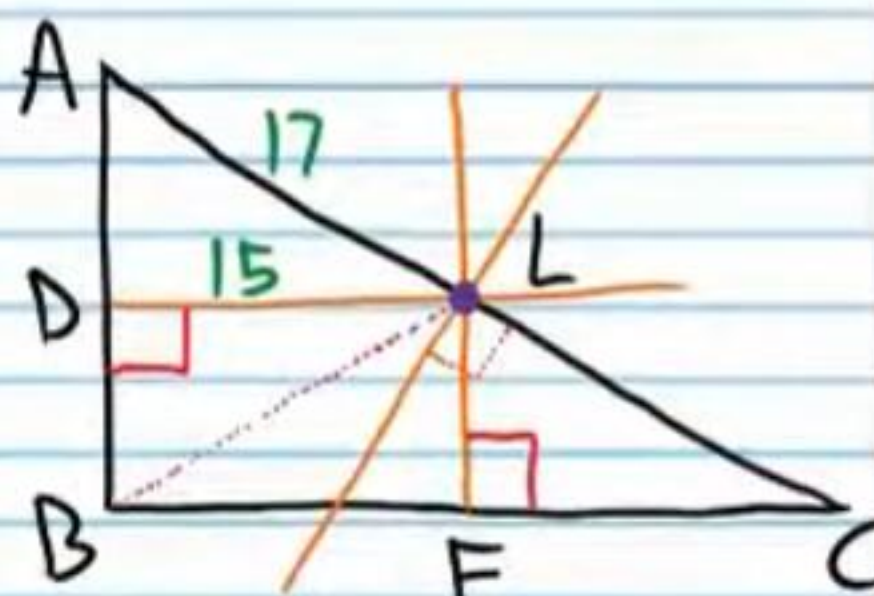
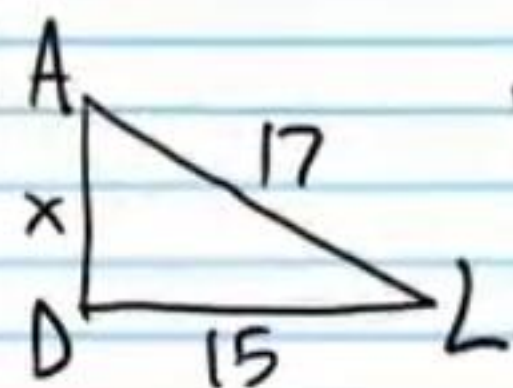
$$x = 5$$

⑧ a) $LC = 17$

b) $BL = 15$

c) $AD = 8$

d) $DB = 8$



$$x^2 + 15^2 = 17^2$$

$$x^2 + 225 = 289$$

$$x^2 = 64$$

$$x = 8$$

If point M is the incenter of each triangle below, find the quantities listed.

⑨

a) $m\angle MBC = 55^\circ$

b) $MJ = 5 \text{ ft}$

c) $m\angle MAJ = 22.6^\circ$

d) $AM = 13 \text{ ft}$

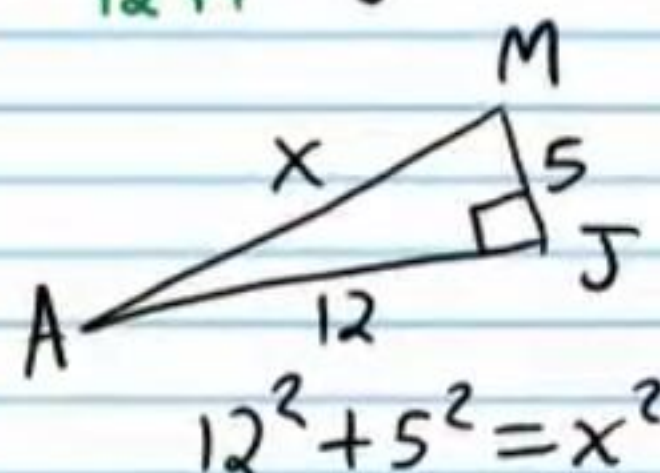
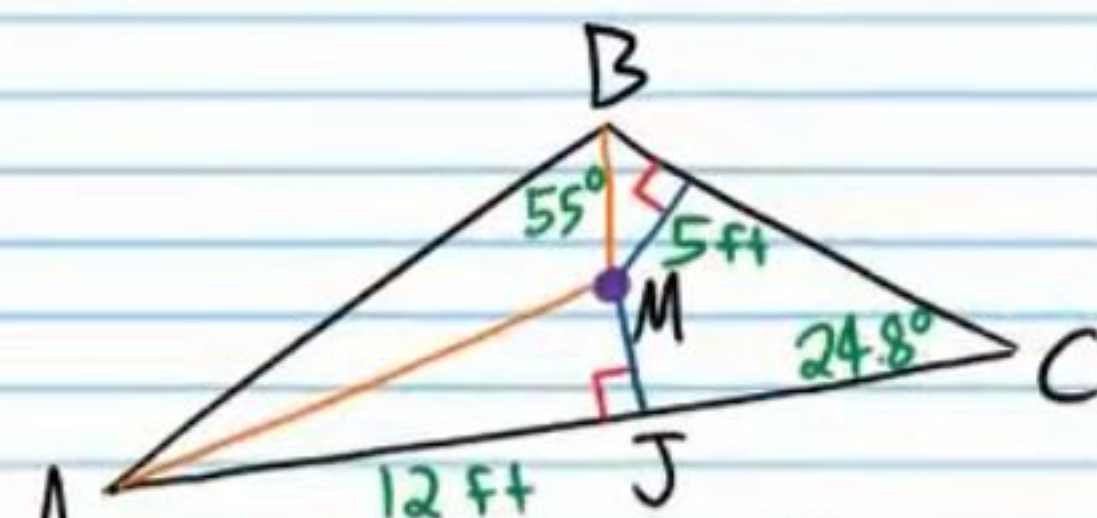
$$m\angle A + m\angle B + m\angle C = 180$$

$$m\angle A + 2(55) + 24.8 = 180$$

$$m\angle A + 134.8 = 180$$

$$m\angle A = 45.2$$

$$m\angle MAJ = 45.2/2 = 22.6^\circ$$



$$12^2 + 5^2 = x^2$$

$$13 = x$$

⑩ a) $m\angle MAC = 37^\circ$

b) $m\angle MCD = 38^\circ$

c) $m\angle DCB = 76^\circ$

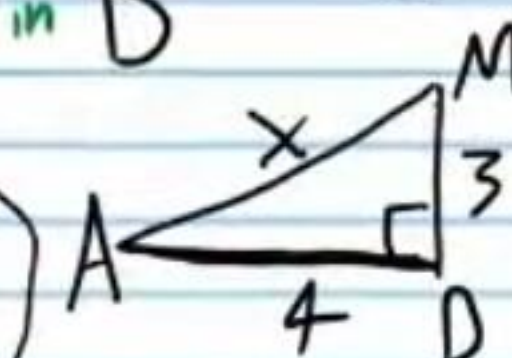
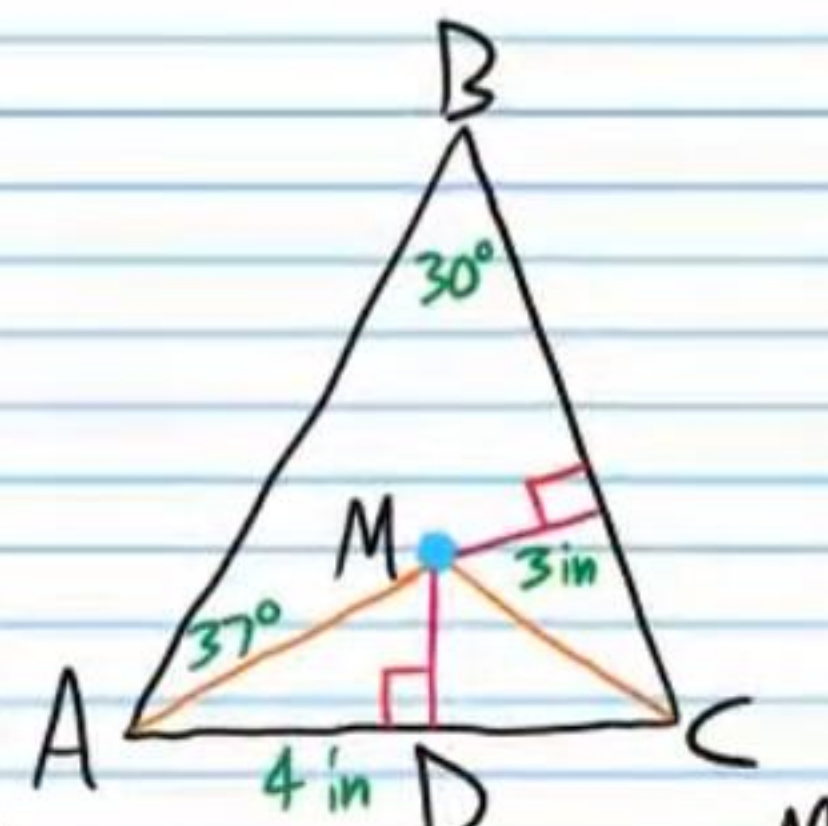
d) $AM = 5 \text{ in}$

$$m\angle A + m\angle B + m\angle C = 180$$

$$2(37) + 30 + m\angle C = 180$$

$$m\angle C = 76^\circ$$

$$m\angle MCD = 76/2 = 38^\circ$$



$$3^2 + 4^2 = x^2$$

$$x = 5$$

⑪ a) $m\angle MBC = 45^\circ$

b) $m\angle MAB = 35^\circ$

c) $m\angle CAB = 70^\circ$

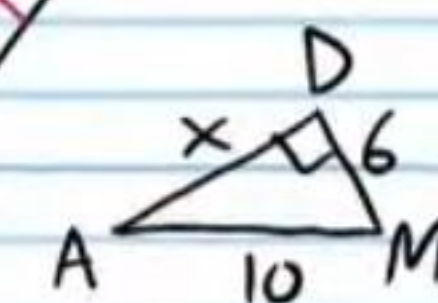
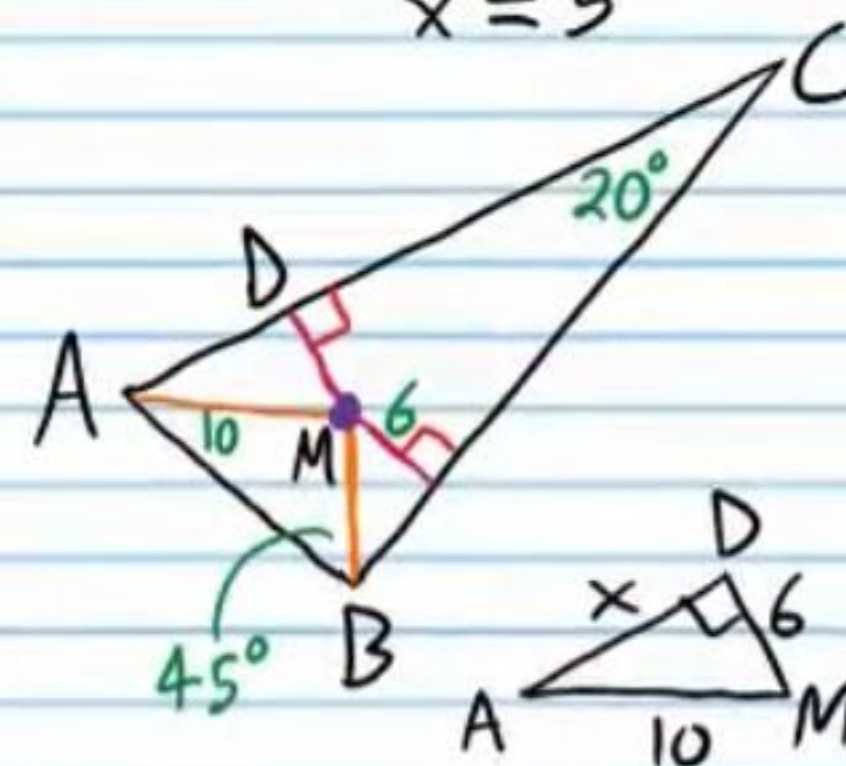
d) $AD = 8$

$$m\angle A + m\angle B + m\angle C = 180$$

$$m\angle A + 45(2) + 20 = 180$$

$$m\angle A = 70^\circ$$

$$m\angle MAB = 70/2 = 35^\circ$$



$$x^2 + 6^2 = 10^2$$

$$x = 8$$

⑫ a) $m\angle BCM = \boxed{37^\circ}$

b) $m\angle ABM = \boxed{23^\circ}$

c) $m\angle ABC = \boxed{46^\circ}$

d) $BM = \boxed{13}$

$m\angle A + m\angle B + m\angle C = 180$

$60 + m\angle B + 37(2) = 180$

$m\angle B = 46^\circ$

$m\angle ABM = \frac{46}{2} = 23^\circ$

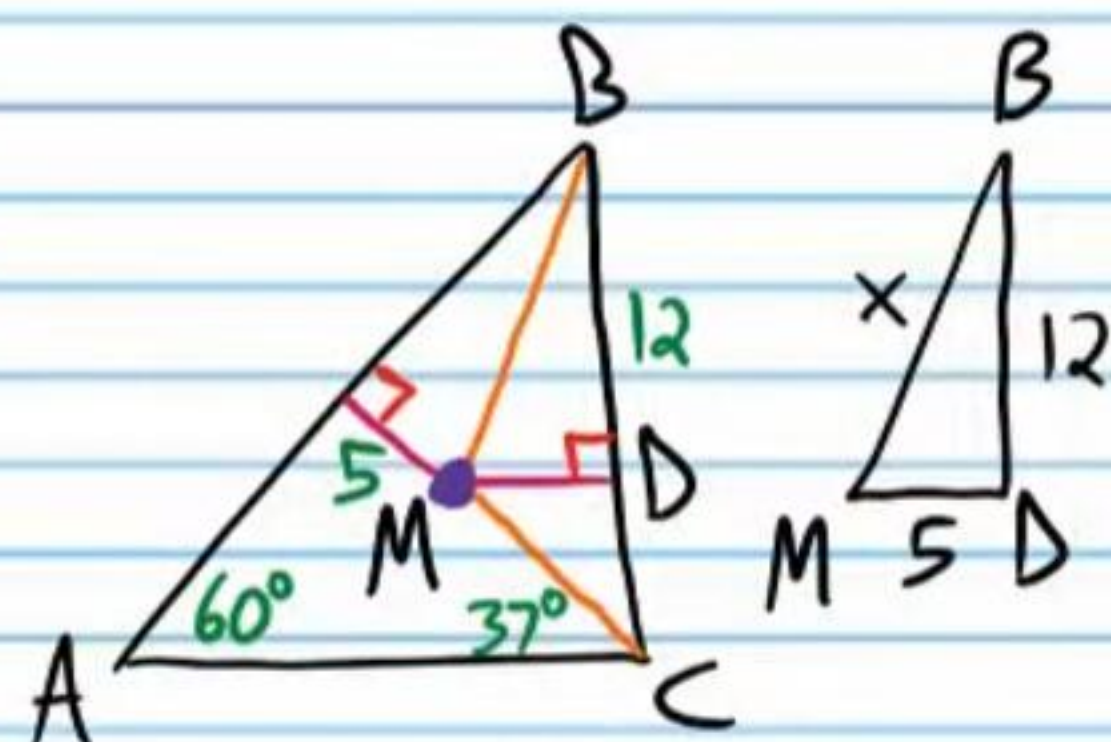
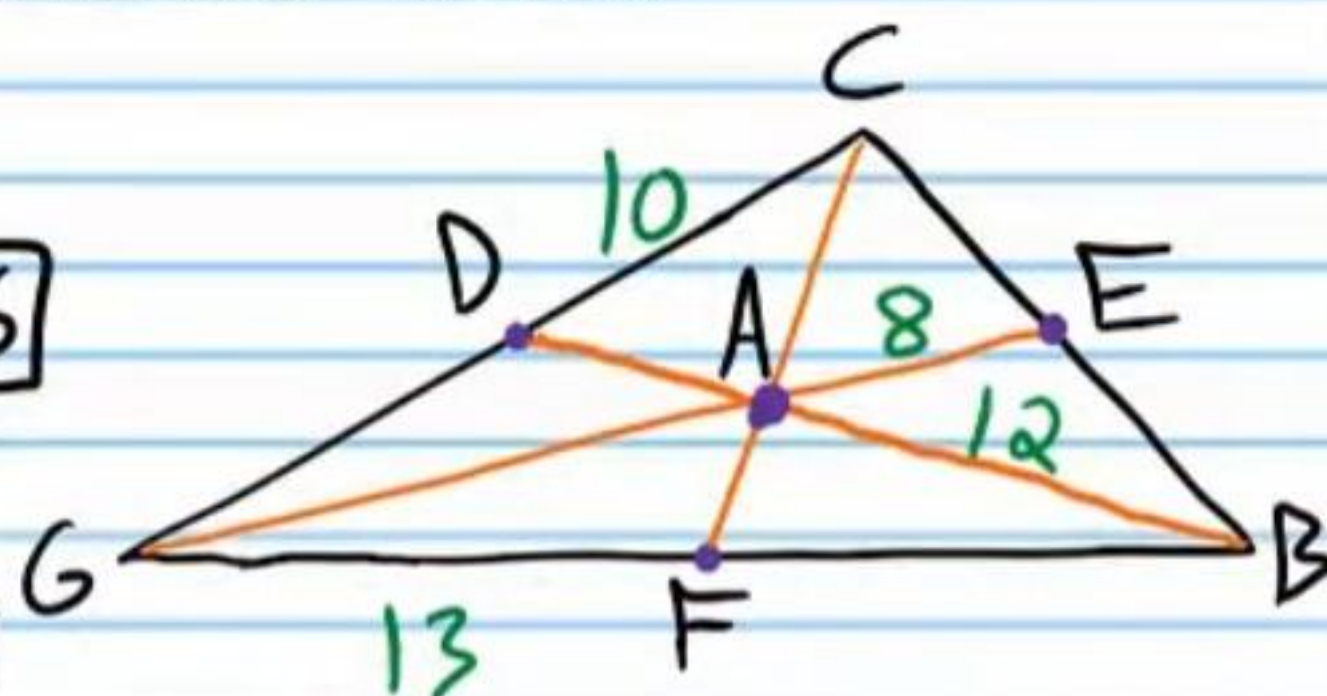
If the segments below are the medians of the triangles, find the quantities listed.

⑬ a) $GD = \boxed{10}$

b) $GA = 2(8) = \boxed{16}$

c) $DA = \frac{12}{2} = \boxed{6}$

d) $GB = 13 + 13 = \boxed{26}$



$x^2 = 12^2 + 5^2$

$x = 13$

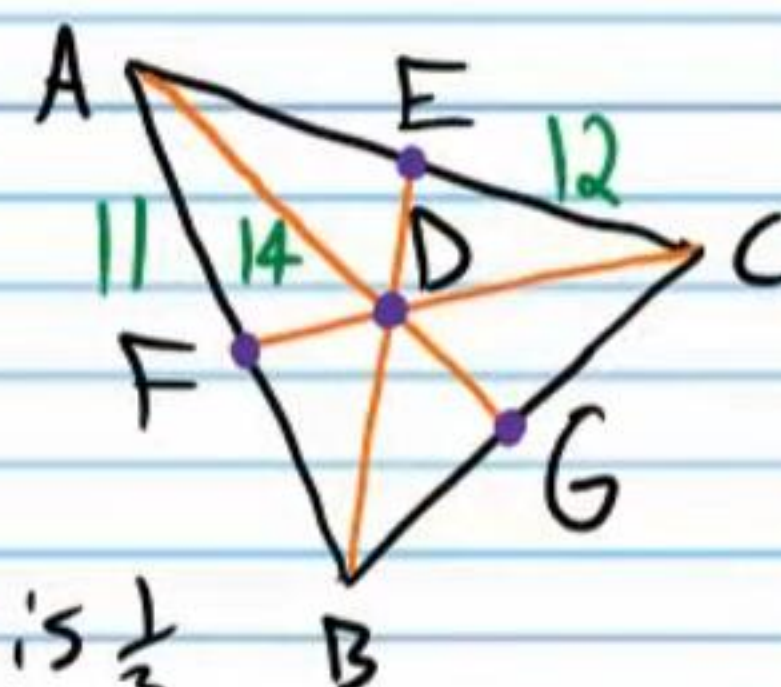
⑭ a) $AE = \boxed{12}$

b) $GD = \frac{14}{2} = \boxed{7}$

c) $AB = 11 + 11 = \boxed{22}$

d) If $FC = 15$
then $DC = 5 + 5 = \boxed{10}$

$\frac{15}{3} = 5$ is $\frac{1}{3}$
the length
of the
median.



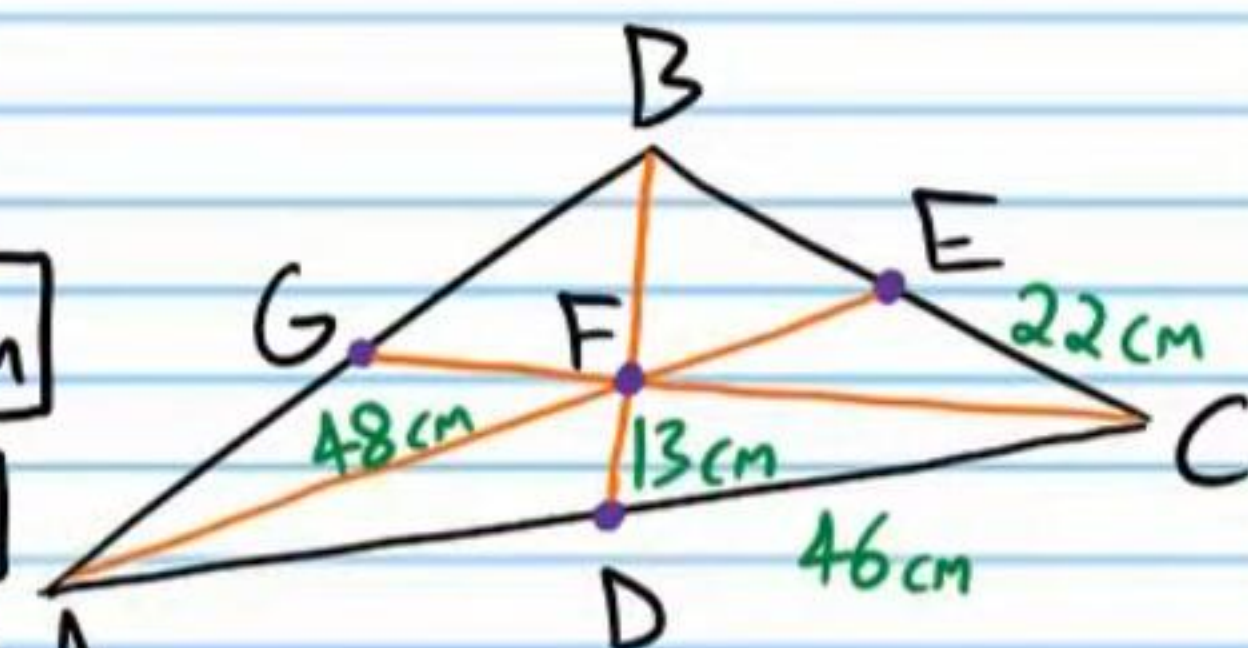
⑮

a) $AD = \boxed{46 \text{ cm}}$

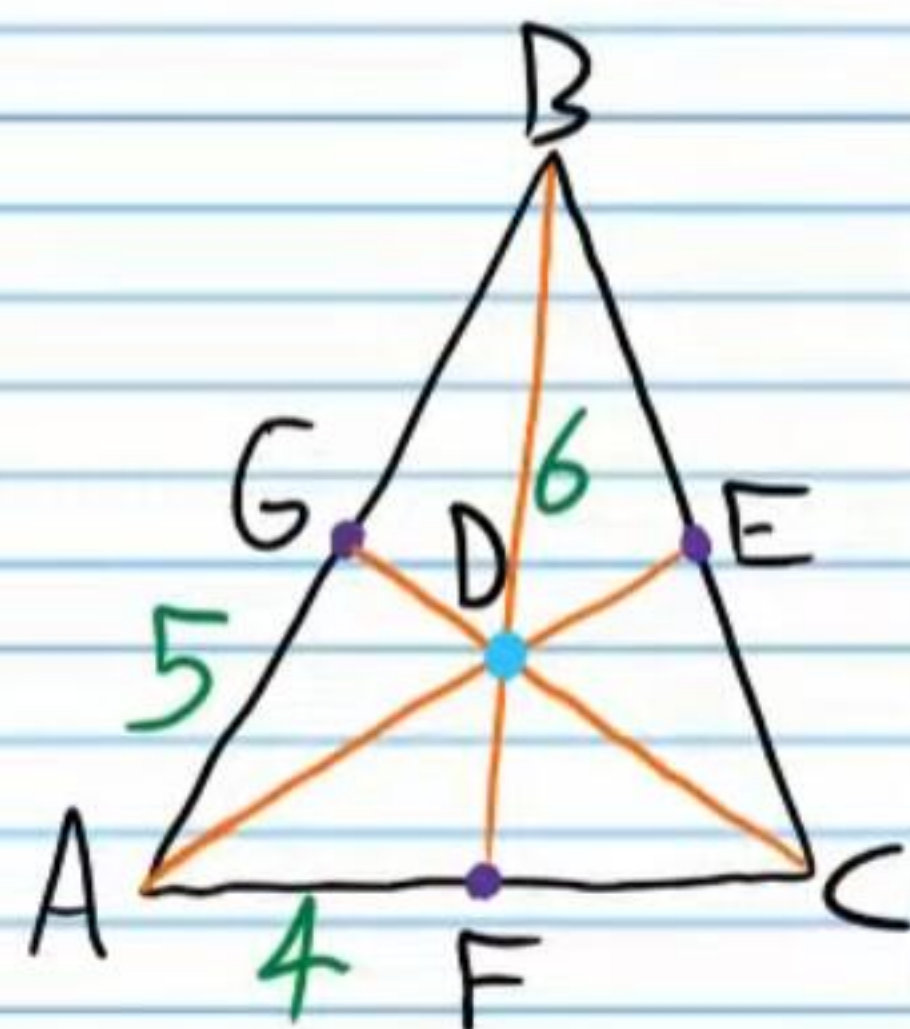
b) $BC = 2(22) = \boxed{44 \text{ cm}}$

c) $BF = 2(13) = \boxed{26 \text{ cm}}$

d) $AE = 48 + 24 = \boxed{72 \text{ cm}}$



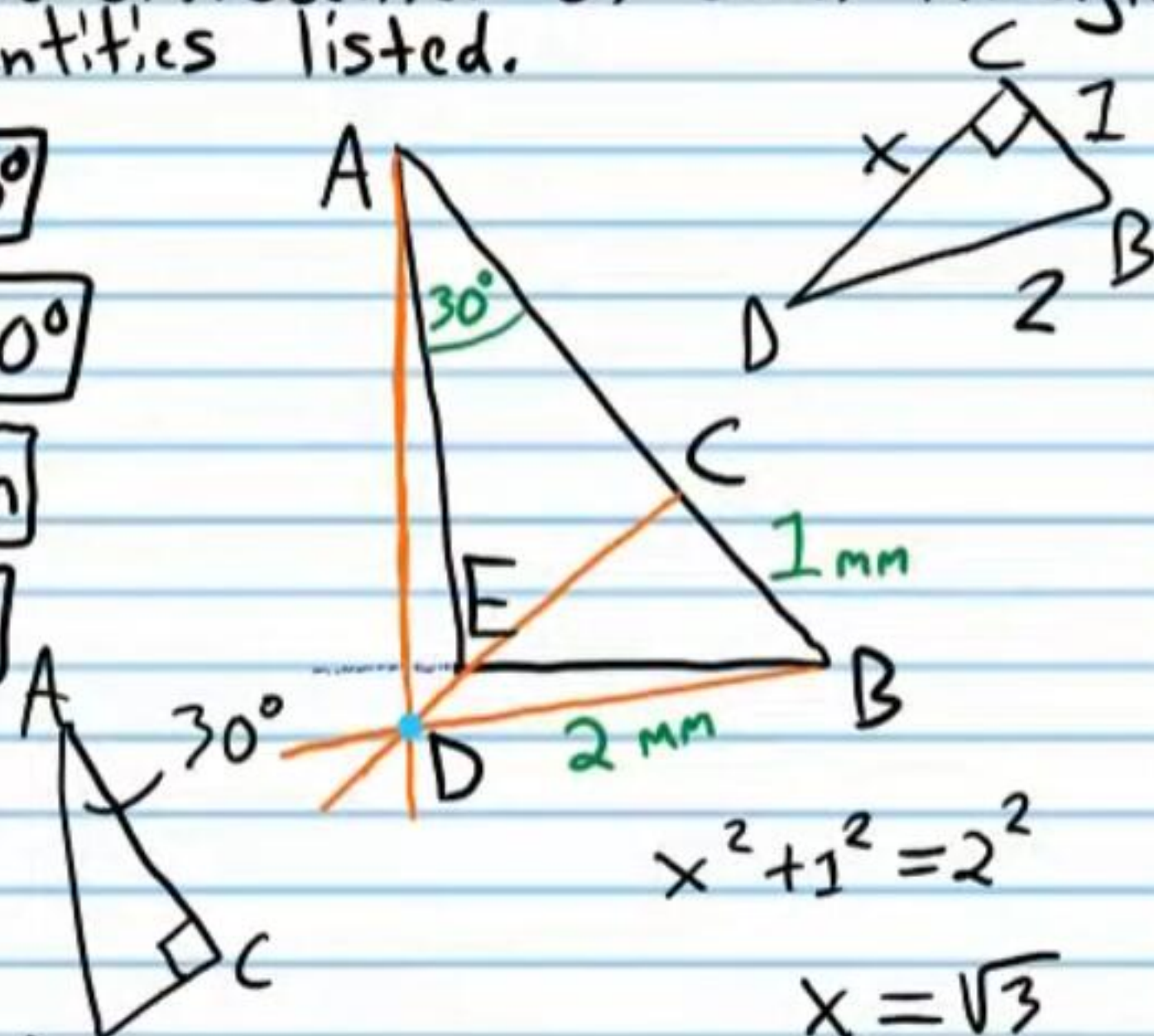
- ⑩ a) $DF = \frac{6}{2} = \boxed{3}$
 b) $AC = 4 + 4 = \boxed{8}$
 c) $GB = \boxed{5}$
 d) If $AE = 6$
 then $DE = \frac{6}{3} = \boxed{2}$



If point D is the orthocenter of each triangle below, find the quantities listed.

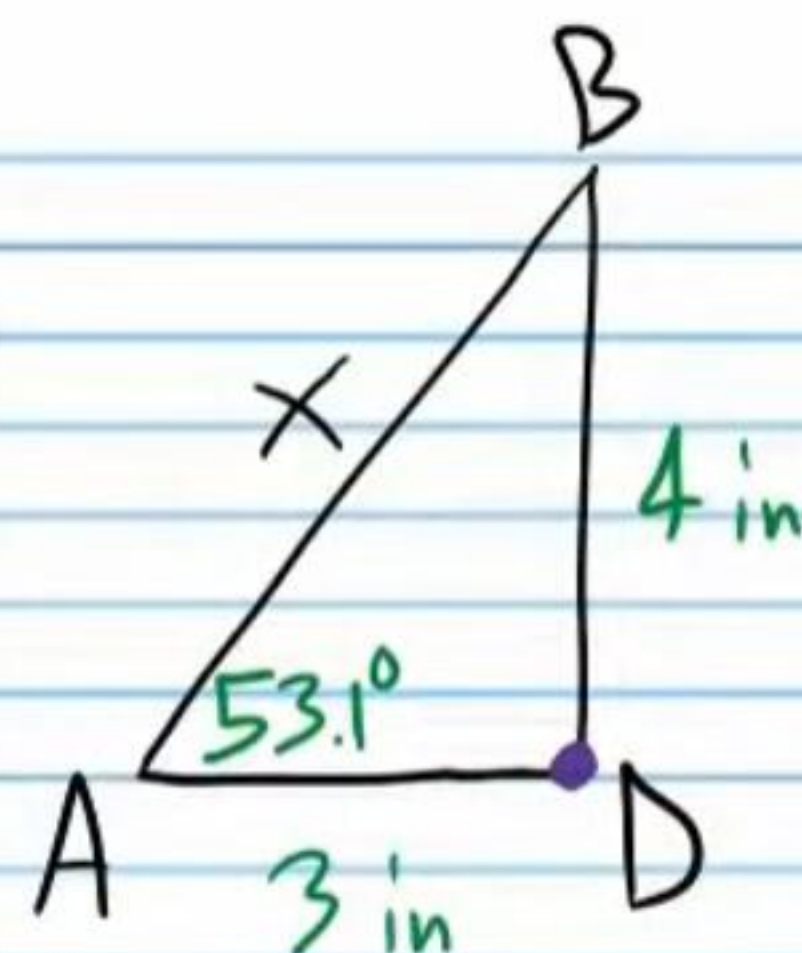
- ⑪ a) $m\angle ACD = \boxed{90^\circ}$
 b) $m\angle AEC = \boxed{60^\circ}$
 c) $CD = \boxed{\sqrt{3} \text{ mm}}$
 d) $m\angle DCB = \boxed{90^\circ}$

$m\angle A + m\angle E + m\angle C = 180$
 $30 + m\angle E + 90 = 180$
 $m\angle E = 60^\circ$



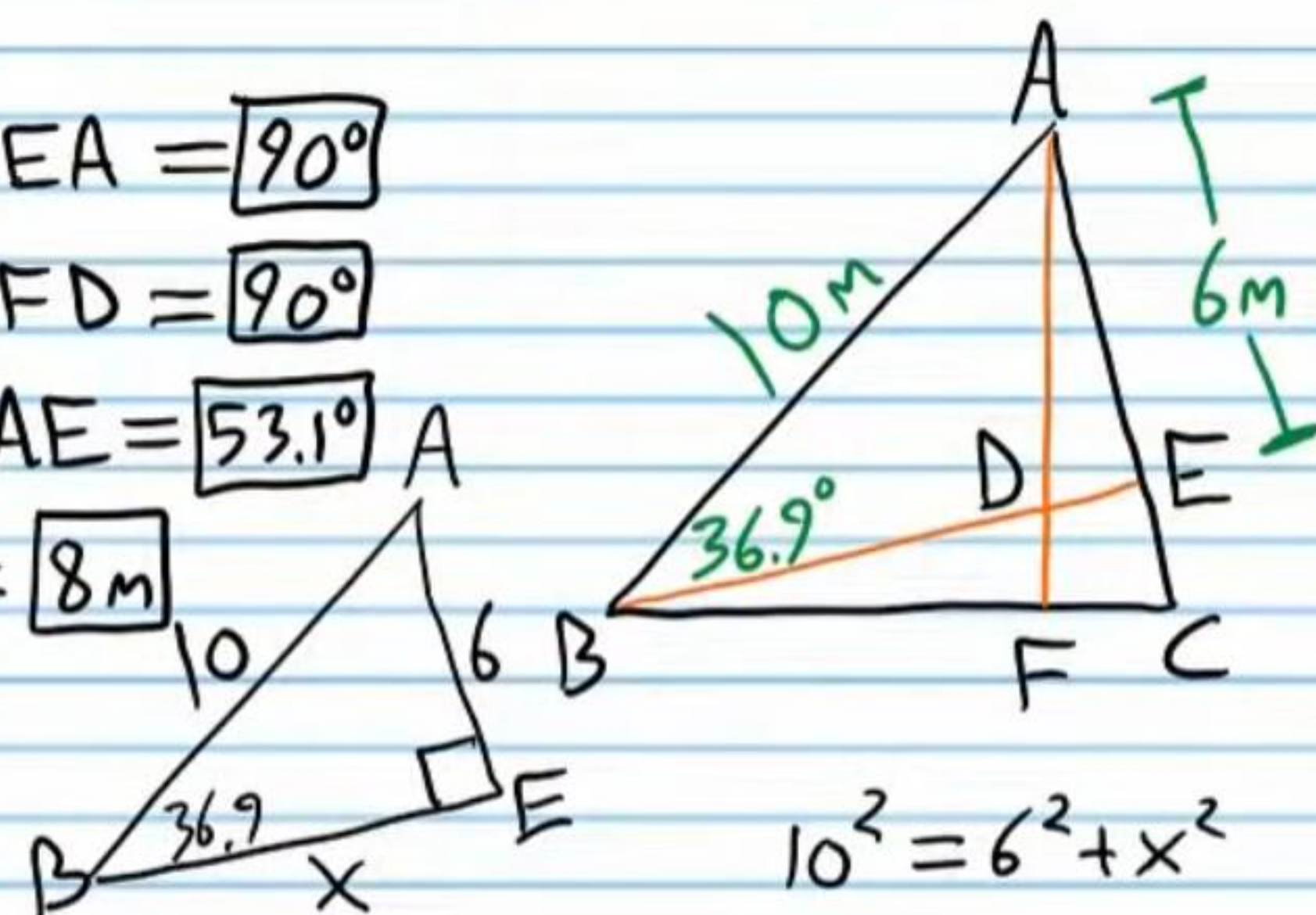
- ⑫ a) $m\angle ADB = \boxed{90^\circ}$
 b) $m\angle ABD = \boxed{36.9^\circ}$
 c) $AB = \boxed{5 \text{ in}}$

$m\angle A + m\angle B + m\angle D = 180$
 $53.1 + m\angle B + 90 = 180$
 $m\angle B = 36.9^\circ$



- ⑬ a) $m\angle DEA = \boxed{90^\circ}$
 b) $m\angle BFD = \boxed{90^\circ}$
 c) $m\angle BAE = \boxed{53.1^\circ}$
 d) $BE = \boxed{8 \text{ m}}$

$m\angle A + m\angle B + m\angle E = 180$
 $m\angle A + 36.9 + 90 = 180$
 $m\angle A = 53.1^\circ$

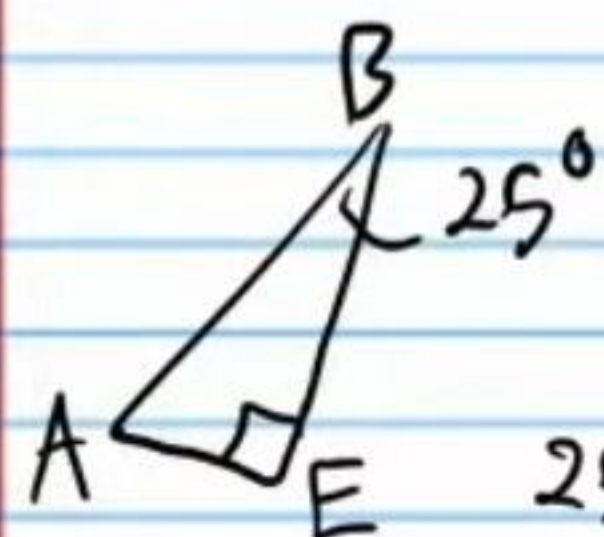


20) a) $m\angle CAE = 65^\circ$

b) $m\angle AFB = 90^\circ$

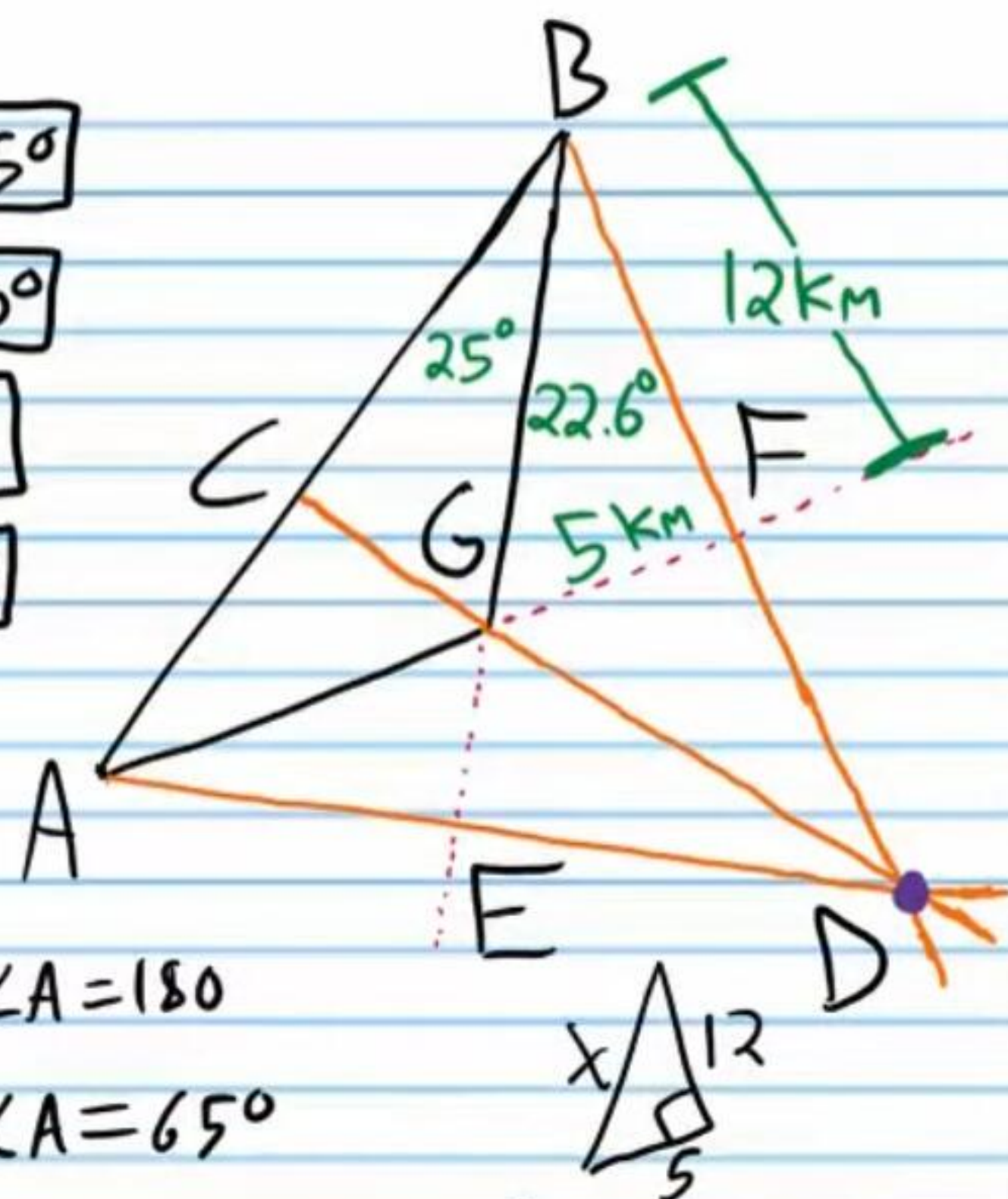
c) $BG = 13 \text{ km}$

d) $m\angle DEB = 90^\circ$



$25 + 90 + m\angle A = 180$

$m\angle A = 65^\circ$



$x^2 = 12^2 + 5^2$
 $x = 13 \text{ km}$

Additional Problems

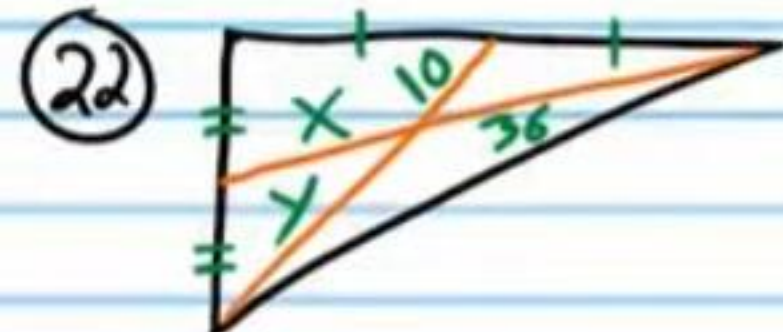
In the following problems, find x and y .



$2x = 10$

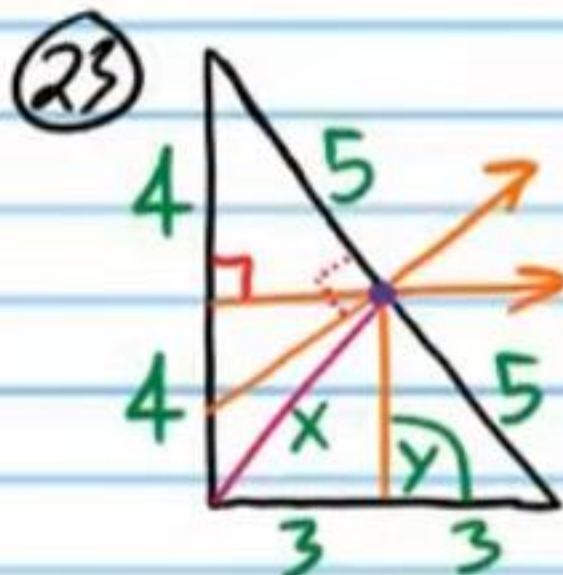
$x = 5$

$y = 38^\circ$



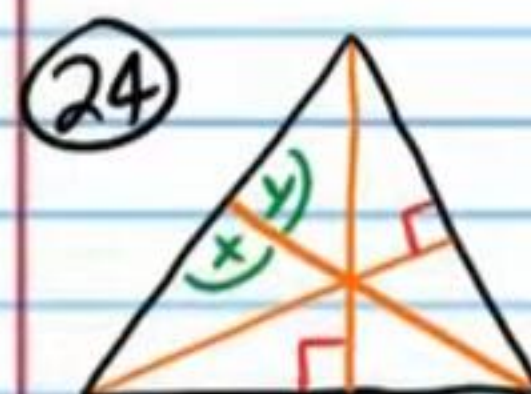
$y = 2(10) = 20$

$x = \frac{36}{2} = 18$



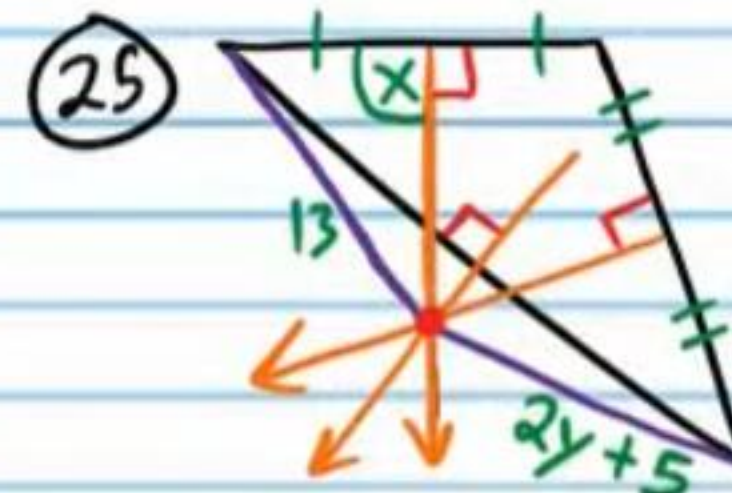
$x = 5$

$y = 90^\circ$



$x = 90^\circ$

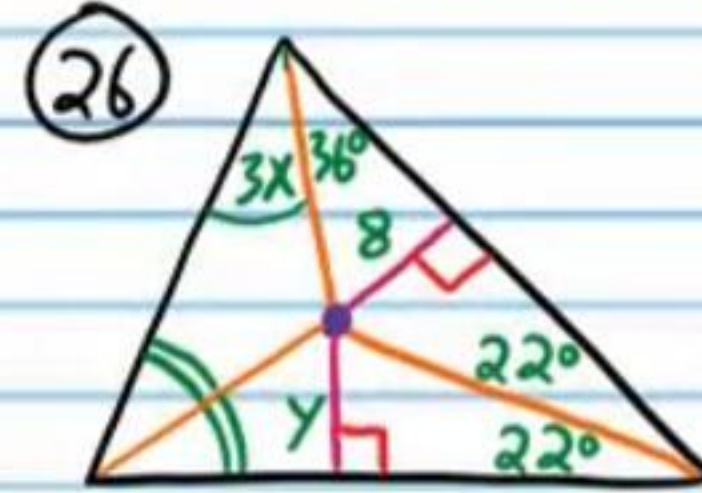
$y = 90^\circ$



$13 = 2y + 5$

$4 = y$

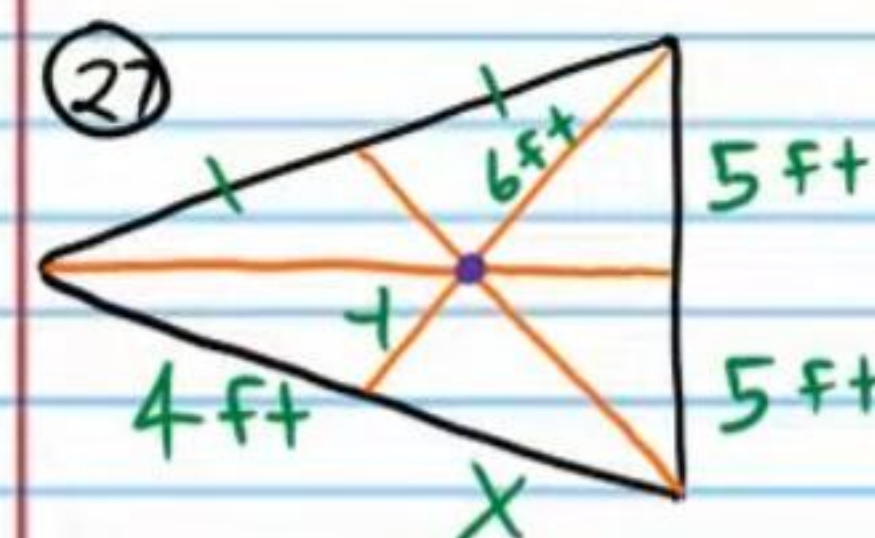
$x = 90^\circ$



$3x = 36$

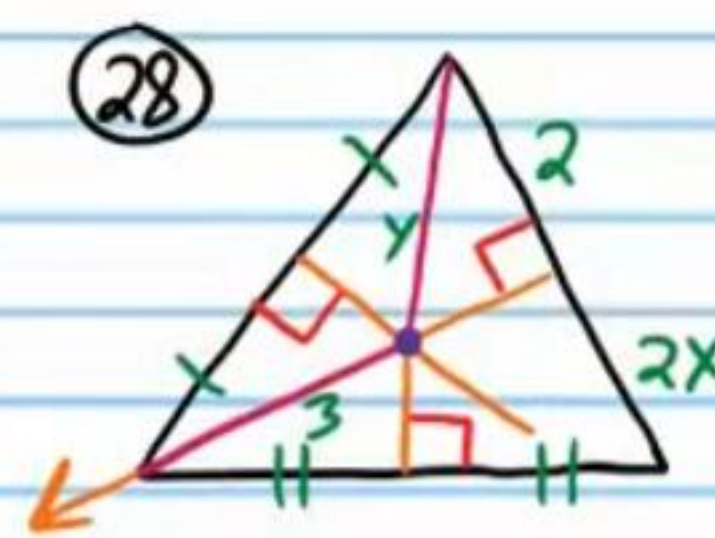
$x = 12$

$y = 8$



$x = 4 \text{ ft}$

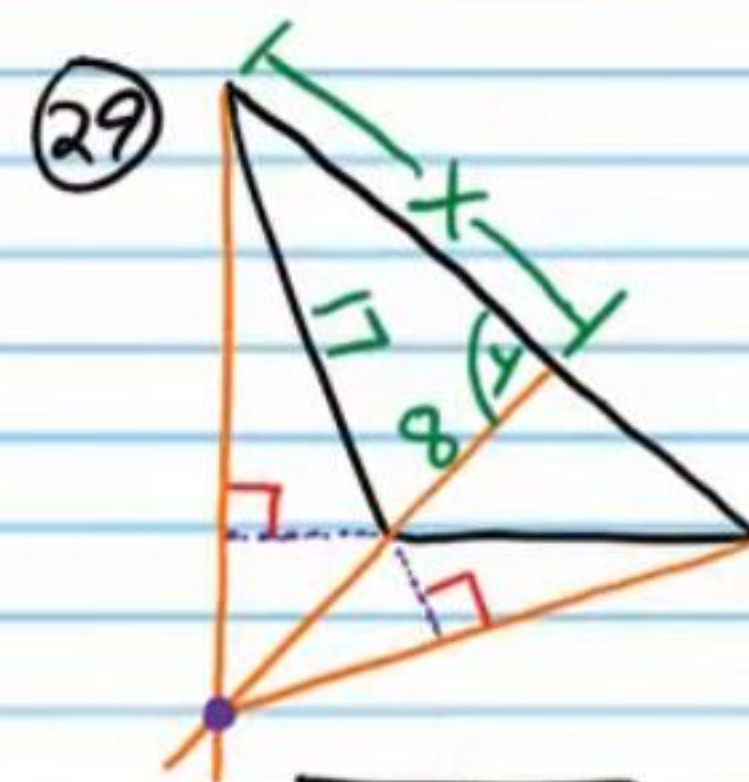
$y = \frac{6}{2} = 3 \text{ ft}$



$2 = 2x$

$1 = x$

$y = 3$



$y = 90^\circ$

$17^2 = 8^2 + x^2$

$289 = 64 + x^2$

$225 = x^2$

$15 = x$